

A system that has to earn the right to name a soldier

What it does, how it works, and why the most important thing about it is what it refuses to do. No technical background needed.

START HERE

The problem, in one paragraph

In the CAPF and the Armed Forces, a jawan under severe stress is usually noticed by a person — a section commander who sees something, or the jawan himself walking in and saying so. Both are late. Between 2018 and 2022, **654 CAPF personnel died by suicide** — roughly one every three days. Another 50,155 resigned or took early retirement in five years.

Meanwhile, the force is already writing things down every day. Who was refused leave. Who has worked twenty-three days without a break. Who has been posted three times in eighteen months. Nobody reads all of it together, because no human could.

IN PLAIN WORDS

The information that would have told us someone was struggling was already in the files. It just wasn't being read.

Why we didn't just build a risk score

The obvious thing to build is a program that reads all those records and gives each person a score out of ten. Anyone can build that. We didn't, and the reason is a number.

The best published research on predicting burnout from personnel records gets it right about **72% of the time**. That sounds reasonable. Now apply it to ten lakh personnel. You have just accused thousands of people, wrongly, of being mentally unwell.

Each one of those is a real soldier being told, in effect, that a computer thinks something is wrong with him. Word travels in a unit. After the first few, nobody fills in a wellness form again, and the whole thing is dead.

THE CORE DECISION

So we stopped letting the computer's guess be the answer. The guess is only allowed to *raise a question*. Something else decides.

HOW IT ACTUALLY WORKS

Four tests, and all four must pass

Think of it like a court, not a scanner. The computer is the person raising a concern — a witness who says "I think something is wrong here." Before that concern becomes a name handed to a welfare officer, it has to survive four tests. Each one is plain arithmetic that a human can check on paper.

1 Is there enough evidence?

Is more than one thing pointing the same way?

Like a court: one witness is not enough, however confident he sounds. This test counts how many different, unrelated records agree — leave, roster, workload, and so on. It is

built so that **no single record can ever pass it**, no matter how extreme. One unusual number about you is not a case about you.

2 Do they agree, or is one shouting?

Is there an ordinary explanation we've missed?

Like a court: before believing the witnesses, check whether they all have the same reason to be wrong. If a jawan's hours went up — but so did everyone's in his unit, because the unit is deployed — then his hours tell you about the deployment, not about him. **This is the test that does most of the work.**

3 Has it lasted?

Is this a pattern, or just a bad week?

Like a court: one incident is not a habit. This test weighs the last month most heavily — long enough that a rough patch doesn't count, short enough to act before things get worse. Everyone has bad weeks. A bad month is different.

4 Is there anything we can do?

And has he agreed to be approached at all?

Like a court: there is no point in a verdict with no remedy. If nobody is already helping him, and there is a real action available — a rest cycle, a leave review, a counselling referral — and **he has consented to being contacted**, this passes. If he hasn't consented, it fails instantly and completely.

Miss any one of the four and no name is produced. Not a weaker warning, not a "possible case" list. Nothing. Nobody learns his name.

Follow it through

Here is the whole system, as it happens to one man. He is not real — this is a run on synthetic records — but every number below came out of the working system.

A jawan in 7th Battalion Head Constable · 11 years' service

Months
before

Things start showing up in the files.

His leave is refused four times. His roster starts showing longer unbroken stretches than he has ever worked. Nobody notices, because these are three different clerks entering three different records in three different systems.

Earlier

He decided what the system may look at.

On his phone, in Hindi, he was shown a screen headed "*what this system can and cannot see about me*" — before being asked to agree to anything. He ticked service records and the wellness questionnaire. He left the fitness-band option off. He also agreed, as a separate question, that a welfare officer may approach him. He could have said no to that and still allowed the analysis.

02:00

The machine runs. Nobody is awake.

It pulls the records, and *immediately replaces every name and service number with a meaningless code*. From this point on, nothing in the system knows who he is. It works out how far each of his numbers has moved **from his own normal** — not from some force-wide average, because a man in a hard posting has always worked hard and that is not news.

02:00

Three assistants argue about him, in parallel.

One builds the strongest case that something is wrong. One hunts for innocent explanations — is the unit deployed? was that leave sanctioned?

are records missing? One looks up what was tried with similar cases before, including the ones that turned out to be nothing.

None of the three gets to decide. They gather facts. If the second one — the one looking for innocent explanations — fails to run, the whole night refuses to name *anybody*.

02:00

The four tests run. This is the only step that decides.

Four sums. Same input, same answer, every time. No cleverness, no model, nothing that can't be checked by hand.

07:00

A welfare officer opens her console.

She sees a line: 240 screened · 128 deviating · 3 escalated.

Three cases. She opens the first.

She is shown three reasons in plain language, the four sums with the actual numbers in them,

and the argument against the flag — placed above the buttons, so she reads it before she can act.

Then a recommended action.

She clicks to reveal his name. The system writes that down — who asked, when, and why — *before* it looks his name up. And she cannot close the case without saying what happened: helped, didn't need help, already being helped, or he declined to talk.

THE CASE THAT PASSED

What "all four passed" looks like

The bar is how far the test got. The black line is the pass mark. Under each is the actual sum — this is what the officer sees, and it is the whole point of the design.

case 1a120eaf...

NAME RELEASED

Enough evidence  **0.754** PASS

$0.50 \times 0.708 + 0.30 \times 0.667 + 0.20 \times 1 = 0.754$

They agree  **1.000** PASS

no innocent explanation found — nothing pushed against it

It has lasted  **0.943** PASS

$0.20 \times 1.000 + 0.45 \times 1.000 + 0.35 \times 0.837 = 0.943$

We can act  **1.000** PASS

action available × not already helped × he consented × capacity = 1.000

His name goes to the welfare officer, with these reasons attached.

Suggested action: referral to the unit counselling cell. Not a diagnosis — the system never says what is wrong with anybody, only that somebody should have a conversation.

THE CASE THAT MATTERED MORE

And what "we are not going to say" looks like

A different man, same night. Three records moving together. Sustained for months. On any ordinary dashboard this is a red flag — the strongest case of the night.

But **61% of his unit shows exactly the same pattern**, because the unit is deployed. The second test notices, and the case collapses by six thousandths of a point.

case pid-mon-0001

NO NAME PRODUCED

Enough evidence  1.000 PASS

three separate records, one of them a form he filled in himself

They agree  0.694 FAILS

needs 0.700 — misses by 0.006. His unit is deployed.

It has lasted  0.965 PASS

breaching nearly every day for three months

We can act  1.000 PASS

everything available, and he had consented

Nobody learns his name. No alert is raised.

Instead the system says what would change its mind, in words the officer can go and check: *"if this is still there after the unit's deployment ends."* And it re-labels the finding as what it actually is — a workload problem for the commanding officer to fix by changing rosters, not a welfare concern about one man.

THIS IS THE WHOLE SYSTEM

It did not stay quiet because someone told it to. It stayed quiet because the sums came out that way — and then it explained exactly what would have changed them.

THE SHAPE OF A NIGHT

How few people reach a human

looked at  240

something changed  128

kept an eye on

125

name released

3

125 people were looked at, found to have changed, and **deliberately not named**. That is not the system failing to find them. That is the system declining to accuse them on evidence that didn't hold up.

WHO SEES WHAT

Four people, four completely different screens

This is enforced deep in the software, not by hiding menu items. If someone opens the wrong page, the system refuses — a mistake in the app could not leak a name, because the name was never sent to it.

The jawan

- ✓ His own records, his own consent, in his own language
- ✓ Why the system looked at him — the same reasons the officer got
- ✗ Anything at all about anybody else

The welfare officer

- ✓ Names, only for cases that passed all four tests, only in her own units
- ✓ The reasons, the sums, and the argument against
- ✗ Anyone the tests refused — no name exists to show her
- ✗ His actual answers to any questionnaire

The commanding officer

- ✓ A heat map of where his unit is running hot, so he can fix rosters
- ✗ Any individual, ever. There is no button, and no way to build one
- ✗ Any square covering fewer than five people — it is removed, not blurred

The auditor

- ✓ Every access, decision and override ever made, in a tamper-proof list
- ✗ Any name. Reading the list never reveals who anyone is
- ✗ The contents of anything — only that an event happened

The commanding officer's screen is the one people ask about most. When a CO asks "so who is it?", the honest answer is a process, not a name: raise the welfare cell's capacity, and let the welfare officer — who is trained for this and is not in his chain of command — work the individual cases.

TWO PROMISES

What it will never be used for

Never for your career

Nothing this system produces can reach your ACR, your promotion, your posting, or any disciplinary process. That isn't a policy in a document — it is blocked in the software for every single role, and tested.

Withdrawing is invisible

If you pull your consent, nobody is told. Not your CO, not your welfare officer. There is no "recently withdrawn" list anywhere for anybody to notice, because we did not build one. Only an auditor can see that an event occurred at all.

Never a diagnosis

The system never says anyone is depressed or unwell. It says somebody should have a conversation, and suggests what kind. A doctor diagnoses. This does not.

If you say it out loud

If you write something urgent into the app — the questionnaire has one question specifically about this — it does not wait for tonight's run and it does not go through the four tests. Someone qualified contacts you the same working day, and it goes to a mental-health authority, never to your commanding officer.

BEING STRAIGHT ABOUT IT

What this cannot do

Every claim above is tested. These are the things we cannot claim.

- **It cannot see a man who hides it well.** Someone who is struggling and says nothing and shows nothing in the records is invisible to this, completely. It is a net with a known mesh size, not a mind-reader.
- **It will still miss people.** We deliberately tuned it to stay quiet when unsure. That choice means some genuine cases wait until the next cycle. We made it on purpose: a missed early signal can be caught next month, and a wrongly named soldier cannot be un-named.
- **Seven of its eight languages have not been checked by a native speaker yet.** They are complete and we wrote them carefully, but the system *refuses to record consent* against an unverified translation. Getting them checked is a real task still outstanding.
- **It has never seen real records.** Everything works, end to end, on realistic invented data. Nothing here has been tested on a real force, and no accuracy figure from this build means anything about real people.

IF YOU REMEMBER ONE THING

The sentence it all comes down to

THE DESIGN IN ONE LINE

The computer may raise the concern. The evidence has to back it up. Plain arithmetic makes the decision. And a human being takes the action.

Everything else in this document is a consequence of putting those four things in that order — and refusing to let the computer skip to the end.

Every figure in this document was read from the working system, not written by hand. 561 automated tests · 8 languages · runs entirely offline with no internet connection.

Smart India Hackathon 2026 · Problem Statement 26186 · Team CodeZila

Welfare support, never disciplinary action.